(1) a) 2 years

b) $q_{25} = 1 - p_{25} = 1 - \frac{1250-10(25)}{1000} = \emptyset$

Not reasonable (25) could die within next year due to health issues or an accident
q_{25} can be small, but should be greater than 0

c) $p_{50} = \frac{1250-507}{1000} = \frac{3}{4}$, $p_{[50]} = \left(\frac{3}{4}\right)^2 = \frac{9}{16} < .75$

Selection indicates lower probability of surviving
Selection indicates worse health

d) $p_{[40]+1}\, p_{42}\, p_{43} = \left(\frac{1250-410}{1000}\right)^{1.5} \frac{830}{1000}(.82) = .523975$

e) $100{,}000\left[v q_{35} + v^2 p_{35} q_{36}\right] = 17{,}153.80$ ($A^1_{35:2\rceil}$)

$p_{35} = .9 \Rightarrow q_{35} = .1$, $p_{36} = .89$, $v = e^{-.1}$

f) $100{,}000^2\left[{}^2A^1_{35:2\rceil} - A^1_{35:2\rceil}\right] = 1{,}188{,}097{,}146$

${}^2A^1_{35:2\rceil} = e^{-.2}(.1) + e^{-.4}(.9)(.11)$

g) PV $+$ EPV has an inverse relation with interest rate

higher $\delta \Rightarrow$ lower EPV

2a) non-profit policy + no expenses used when calculating the premium ⇒ net premium

$P\ddot{a}_{[35]:\overline{10}} = 200{,}000\, A_{[35]:\overline{25}}$

$= 200{,}000\,(.24198)$

$\ddot{a}_{[35]:\overline{10}} = \ddot{a}_{35:\overline{10}} - v^{10}\,{}_{10}p_{[35]}\,\ddot{a}_{45}$

$= 15.993 - 1.06^{-10}\left(\frac{9801.3123}{9892.9157}\right)(14.85) = 7.777618$

$P = 6222.47$

b) $V(10) = 200{,}000\, A_{45:\overline{15}} = 200{,}000(.42556) = 85{,}112$

c) $85{,}112 - 150 = Y\,\bar{A}_{45:\overline{20}}$ Assume Claims Acceleration or UDD

$\bar{A}_{45:\overline{20}} = 1.06^{1/2} A'_{45:\overline{20}} + A^{1}_{45:\overline{20}}$

$1.06^{1/2}(A_{45} - 1.06^{-20}\,{}_{20}p_{45}\,A_{65}) + 1.06^{-20}\,{}_{20}p_{45} = .328688$

$A_{45} = .15443$, $A_{65} = .40177$ ${}_{20}p_{45} = \frac{l_{65}}{l_{45}} = \frac{8821.2612}{9801.3123} = .9000$

$Y = 258{,}488.33$

d) current observed / actual int. > expected basis int would allow BOY funds to be invested + exceed EOY obligations ⇒ profit

mort basis > actual mortality is another option

(3) a) $Z = 10{,}000\, \ddot{a}_{\overline{K_{[70]}+1|}}$

b) $Var(Z) = 10{,}000^2 Var\left(\frac{1-v^{K_{[70]}+1}}{d}\right) = \frac{10{,}000^2}{d^2}\left({}^2A_{[70]} - A_{[70]}^2\right)$

$= 1{,}182{,}607{,}574 \quad \left\{ d = \frac{.06}{1.06} \right.$ $\quad {}^2A_{[70]} = .26583$ $\quad A_{[70]} = .47743$

c) $Pr\left(\frac{10{,}000}{10{,}000}\frac{1-v^{K_{[70]}+1}}{d} \geq \frac{88{,}888}{10{,}000}\right) =$

$Pr\left(1 - v^{K_{[70]}+1} \geq 8.8888d\right) =$

$Pr\left(1 - 8.8888d \geq v^{K_{[70]}+1}\right) =$ $\quad e^{-\delta(K_{[70]}+1)}$

$Pr\left[\ln(1-8.8888d) \geq -\delta(K_{[70]}+1)\right] =$

$\delta = \ln(1.06)$

$Pr\left[11.003 \leq K_{[70]}\right] = {}_{12}p_{[70]} =$

$\frac{4527.496}{7960.9776} = .5687$

d) $Var(Y) = Var(Z)$, only difference is a certain payment at time 0, no impact on the variance

(4) a) $\frac{l_{72.6} - l_{76.6}}{l_{70.3}} = \frac{.4l_{72} + .6l_{73} - (.4l_{76} + .6l_{77})}{.7l_{70} + .3l_{71}}$

$= .1195$

b) Prob. that (70.3) lives 2.3 years + then dies in next 4 years

c) ${}_{.3}p_{40.7}\,({}_{14}p_{41}) = p_{40}^{.3}\ \frac{94,532}{97847} = .96581$

$p_{40} = \frac{97847}{97,952}$

d) ${}_{.2}q_{40} = .000214$

${}_{.2}p_{40} = 1 - {}_{.2}q_{40} = .999786$

$F = F' = (tq_x)' = q_x$

${}_tp_x \mu_{x+t}$

$\mu_{40.2} = \frac{q_{40}}{{}_{.2}p_{40}} = \frac{.00107}{.999786}$ ✓

DB+exp

(5a) $V(14)(1.06) = 10{,}100(.01) + .99\,V(15)$

q p

$V(15) = 5786.87$

b) profit > 0 because

actual death benefit < expected death benefit

$2 \times 10{,}000 < 80(.03)\,10{,}000$

c) interest equals expected claim expense amount, ~~expected~~ but expense is earlier than expected. Hence, profit < 0

higher than expected (£240 vs £200) and